

SOLIDWORKS API (.NET VBA C#) — Reference

(41/337): ICircularPatternFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ICircularPatternFeatureData (Type)

ICircularPatternFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICircularPatternFeatureData Interface

Allows access to a circular pattern feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ICircularPatternFeatureData
```

Visual Basic (Usage) 

```
Dim instance As ICircularPatternFeatureData
```

C#

```
public interface ICircularPatternFeatureData
```

C++/CLI

```
public interface class ICircularPatternFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData](#).

Example

[Get Transform for Each Circular Pattern Instance \(C#\)](#)
[Get Transform for Each Circular Pattern Instance \(VB.NET\)](#)
[Get Transform for Each Circular Pattern Instance \(VBA\)](#)
[Create Bidirectional Circular Pattern Feature \(C#\)](#)
[Create Bidirectional Circular Pattern Feature \(VB.NET\)](#)
[Create Bidirectional Circular Pattern Feature \(VBA\)](#)

Remarks

Read [Pattern Features and their Feature Data Objects](#).
Before calling IFeatureManager::CreateDefinition, you must pre-select the entities needed to create the circular pattern feature.

If...	Use...
Using IModelDocExtension::SelectByID2 to select entities	• 1 to mark the direction axis • 4 to mark the features to pattern • 256 to mark the bodies to pattern • 134217728 to mark the structure systems to pattern

You must call [IFeatureManager::CreateFeature](#) after initializing this feature data object in order to successfully create the circular pattern feature

After creating the circular pattern feature, you can:

- [vary the spacing of pattern instances](#) and dimensions by implementing [IInstanceToVaryOptions](#).
- [change the structure system to pattern](#).

Accessors

[IFeature::GetDefinition](#)
[IFeatureManager::CreateDefinition](#)

Access Diagram

See Also

[ICircularPatternFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCircularPatternFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICircularPatternFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ICircularPatternFeatureData Interface

The following tables list the members exposed by [ICircularPatternFeatureData](#).

Public Properties

Name	Description
 AutoSelect	Gets whether to automatically select all bodies in a multibody part intersected by this circular pattern feature.
 Axis	Gets or sets the entity used to determine the direction of this circular pattern feature.
 BodyPattern	Gets or sets whether to base this circular pattern feature on bodies and structure systems or features and faces.
 Direction2	Gets or sets whether to create a bidirectional circular pattern feature.
 EqualSpacing	Gets or sets whether the pattern instances in Direction 1 are equally spaced in this circular pattern feature.
 EqualSpacing2	Gets or sets whether the pattern instances in Direction 2 are equally spaced in this bidirectional circular pattern feature.
 FeatureScope	Gets which bodies in this multibody part are affected by this circular pattern feature.
 FeatureScopeBodies	Gets the bodies in this multibody part that are affected by this circular pattern feature.
 GeometryPattern	Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.
 InstancesToVary	Gets or sets whether to vary the spacing of pattern instances in this circular pattern feature.
 PatternBodyArray	Gets or sets a list of bodies to pattern in a multibody part for this circular pattern feature.
 PatternElement	Gets or sets the type of entities on which to base this circular pattern feature.
 PatternFaceArray	Gets or sets the list of faces to pattern for this circular pattern feature.
 PatternFeatureArray	Gets or sets the seed features for the circular pattern feature.
 PropagateVisualProperty	Gets or sets whether to propagate visual properties (e.g., colors, textures, cosmetic thread data, etc.) to all pattern instances in this circular pattern feature.
 ReverseDirection	Gets or sets whether the direction of the axis should be reversed in this circular pattern feature.
 SkippedItemArray	Gets or sets the list of items to skip in this circular pattern feature.
 Spacing	Gets or sets the spacing between pattern instances in Direction 1 of the circular pattern feature.
 Spacing2	Gets or sets the spacing between pattern instances in Direction 2 of this bidirectional circular pattern feature.
 StructureSystemToPatternArray	Gets or sets the structure systems to pattern in this circular pattern feature.
 Symmetric	Gets or sets whether to create a symmetric or asymmetric circular pattern feature in Direction 2.
 TotalInstances	Gets or sets the total number of pattern instances in Direction 1 of this circular pattern feature.
 TotalInstances2	Gets or sets the total number of pattern instances in Direction 2 of this bidirectional circular pattern feature.
 VarySketch	Gets or sets whether to allow pattern instances of a seed sketch to vary in relation to the base part in this circular pattern.

[Top](#)

Public Methods

Name	Description
 AccessSelections	Gains access to selections used to define a circular pattern feature.
 GetAxisType	Gets the type of geometry used to determine the direction of this circular pattern.
 GetInstanceToVaryOptions	Gets the options for varying the spacing of pattern instances in this circular pattern feature.

	GetPatternBodyCount	Gets the number of seed bodies in this circular pattern feature.
	GetPatternFaceCount	Gets the number of patterned faces.
	GetPatternFeatureCount	Gets the number of seed features used to create this circular pattern.
	GetSkippedItemCount	Gets the number of items skipped in this circular pattern.
	GetTransform	Gets the transform for the specified instance of this circular-pattern feature.
	IAccessSelections	Obsolete. Superseded by ICircularPatternFeatureData.IAccessSelections2 .
	IAccessSelections2	Gains access to selections used to define a circular pattern feature.
	IGetPatternBodyArray	Gets a list of the seed bodies for this circular pattern.
	IGetPatternFaceArray	Gets a list of patterned faces for this circular pattern.
	IGetPatternFeatureArray	Gets the seed features for this circular pattern.
	IGetSkippedItemArray	Gets an array of integers that represent the skipped items in this circular feature.
	ISetPatternBodyArray	Sets a list of seed bodies for the circular pattern.
	ISetPatternFaceArray	Sets a list of patterned faces for this circular pattern.
	ISetPatternFeatureArray	Sets the seed features to use to create the circular pattern.
	ISetSkippedItemArray	Sets the list of skipped items in this circular pattern.
	ReleaseSelectionAccess	Releases access to the selections that define this circular pattern.
	SetFeatureScope	Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this circular pattern feature.
	SetInstanceToVaryOptions	Sets the options for varying the spacing of pattern instances in this circular pattern feature.

[Top](#)

See Also

[ICircularPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCircularPatternFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICircularPatternFeatureData Interface Methods

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : [ICircularPatternFeatureData Interface](#)

For a list of all members of this type, see [ICircularPatternFeatureData members](#).

Public Methods

Name	Description
	AccessSelections Gains access to selections used to define a circular pattern feature.
	GetAxisType Gets the type of geometry used to determine the direction of this circular pattern.
	GetInstanceToVaryOptions Gets the options for varying the spacing of pattern instances in this circular pattern feature.
	GetPatternBodyCount Gets the number of seed bodies in this circular pattern feature.
	GetPatternFaceCount Gets the number of patterned faces.
	GetPatternFeatureCount Gets the number of seed features used to create this circular pattern.
	GetSkippedItemCount Gets the number of items skipped in this circular pattern.
	GetTransform Gets the transform for the specified instance of this circular-pattern feature.
	IAccessSelections Obsolete. Superseded by ICircularPatternFeatureData.IAccessSelections2 .
	IAccessSelections2 Gains access to selections used to define a circular pattern feature.
	IGetPatternBodyArray Gets a list of the seed bodies for this circular pattern.
	IGetPatternFaceArray Gets a list of patterned faces for this circular pattern.
	IGetPatternFeatureArray Gets the seed features for this circular pattern.

	IGetSkippedItemArray	Gets an array of integers that represent the skipped items in this circular feature.
	ISetPatternBodyArray	Sets a list of seed bodies for the circular pattern.
	ISetPatternFaceArray	Sets a list of patterned faces for this circular pattern.
	ISetPatternFeatureArray	Sets the seed features to use to create the circular pattern.
	ISetSkippedItemArray	Sets the list of skipped items in this circular pattern.
	ReleaseSelectionAccess	Releases access to the selections that define this circular pattern.
	SetFeatureScope	Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this circular pattern feature.
	SetInstanceToVaryOptions	Sets the options for varying the spacing of pattern instances in this circular pattern feature.

[Top](#)

See Also

[ICircularPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCircularPatternFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICircularPatternFeatureData Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ICircularPatternFeatureData Interface

For a list of all members of this type, see [ICircularPatternFeatureData members](#).

Public Properties

Name	Description
 AutoSelect	Gets whether to automatically select all bodies in a multibody part intersected by this circular pattern feature.
 Axis	Gets or sets the entity used to determine the direction of this circular pattern feature.
 BodyPattern	Gets or sets whether to base this circular pattern feature on bodies and structure systems or features and faces.
 Direction2	Gets or sets whether to create a bidirectional circular pattern feature.
 EqualSpacing	Gets or sets whether the pattern instances in Direction 1 are equally spaced in this circular pattern feature.
 EqualSpacing2	Gets or sets whether the pattern instances in Direction 2 are equally spaced in this bidirectional circular pattern feature.
 FeatureScope	Gets which bodies in this multibody part are affected by this circular pattern feature.
 FeatureScopeBodies	Gets the bodies in this multibody part that are affected by this circular pattern feature.
 GeometryPattern	Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.
 InstancesToVary	Gets or sets whether to vary the spacing of pattern instances in this circular pattern feature.
 PatternBodyArray	Gets or sets a list of bodies to pattern in a multibody part for this circular pattern feature.
 PatternElement	Gets or sets the type of entities on which to base this circular pattern feature.
 PatternFaceArray	Gets or sets the list of faces to pattern for this circular pattern feature.
 PatternFeatureArray	Gets or sets the seed features for the circular pattern feature.
 PropagateVisualProperty	Gets or sets whether to propagate visual properties (e.g., colors, textures, cosmetic thread data, etc.) to all pattern instances in this circular pattern feature.
 ReverseDirection	Gets or sets whether the direction of the axis should be reversed in this circular pattern feature.
 SkippedItemArray	Gets or sets the list of items to skip in this circular pattern feature.
 Spacing	Gets or sets the spacing between pattern instances in Direction 1 of the circular pattern feature.
 Spacing2	Gets or sets the spacing between pattern instances in Direction 2 of this bidirectional circular pattern feature.
 StructureSystemToPatternArray	Gets or sets the structure systems to pattern in this circular pattern feature.
 Symmetric	Gets or sets whether to create a symmetric or asymmetric circular pattern feature in Direction 2.
 TotalInstances	Gets or sets the total number of pattern instances in Direction 1 of this circular pattern feature.
 TotalInstances2	Gets or sets the total number of pattern instances in Direction 2 of this bidirectional circular pattern feature.



Gets or sets whether to allow pattern instances of a seed sketch to vary in relation to the base part in this circular pattern.

[Top](#)

See Also

[ICircularPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCircularPatternFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AccessSelections Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : AccessSelections Method (ICircularPatternFeatureData)

Gains access to selections used to define a circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections(  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim TopDoc As System.Object  
Dim Component As System.Object  
Dim value As System.Boolean  
  
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(  
    System.object TopDoc,  
    System.object Component  
)
```

C++/CLI

```
System.bool AccessSelections(  
& System.Object^ TopDoc,  
& System.Object^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc
Top-level document

Component
Component for the feature

Return Value

True if the selections are successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::AccessSelections](#).

Remarks

Remark

To modify a feature in a...

Then...

Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [ICircularPatternFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::IAccessSelections2 Method](#)

Availability

SOLIDWORKS 2001 FCS Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

AutoSelect Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : AutoSelect Property (ICircularPatternFeatureData)

Gets whether to automatically select all bodies in a multibody part intersected by this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property AutoSelect As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean

value = instance.AutoSelect
```

C#

```
System.bool AutoSelect {get;}
```

C++/CLI

```
property System.bool AutoSelect {
    System.bool get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to automatically select all bodies intersected by this circular pattern feature, false to specify affected bodies

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::AutoSelect](#).

Remarks

This property is valid only if [ICircularPatternFeatureData::GeometryPattern](#) is true.

Call [ICircularPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Circular Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Axis Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : Axis Property (ICircularPatternFeatureData)

Gets or sets the entity used to determine the direction of this circular pattern feature.

Visual Basic Syntax

Visual Basic (Declaration)

Property Axis As System.Object

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object
```

```
instance.Axis = value
```

```
value = instance.Axis
```

C#

```
System.object Axis {get; set;}
```

C++/CLI

```
property System.Object^ Axis {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Direction entity (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::Axis](#).

Remarks

This property returns Nothing or null if you do not call [ICircularPatternFeatureData::AccessSelections](#) before using this property.

If [ICircularPatternFeatureData::GetAxisType](#) returns reference axis, use [IFeature::GetSpecificFeature2](#) to get [IRefAxis](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::VarySketch Property \(\)](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BodyPattern Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : BodyPattern Property (ICircularPatternFeatureData)

Gets or sets whether to base this circular pattern feature on bodies and structure systems or features and faces.

.NET Syntax

Visual Basic (Declaration)

```
Property BodyPattern As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean

instance.BodyPattern = value

value = instance.BodyPattern
```

C#

```
System.bool BodyPattern {get; set;}
```

C++/CLI

```
property System.bool BodyPattern {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True for bodies and structure systems, false for features and faces

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::BodyPattern](#).

Remarks

If this property is:

- True, use [ICircularPatternFeatureData::PatternBodyArray](#) to specify the bodies to pattern and [ICircularPatternFeatureData::StructureSystemToPatternArray](#) to specify the structure systems to pattern.
- False, use [ICircularPatternFeatureData::PatternFaceArray](#) or [ICircularPatternFeatureData::PatternFeatureArray](#) to specify the face or feature to pattern.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::PatternElement Property \(\)](#)
[ICircularPatternFeatureData::IGetPatternBodyArray Method \(\)](#)
[ICircularPatternFeatureData::IGetPatternFaceArray Method \(\)](#)
[ICircularPatternFeatureData::IGetPatternFeatureArray Method \(\)](#)
[ICircularPatternFeatureData::ISetPatternBodyArray Method \(\)](#)
[ICircularPatternFeatureData::ISetPatternFaceArray Method \(\)](#)
[ICircularPatternFeatureData::ISetPatternFeatureArray Method \(\)](#)

Availability

SOLIDWORKS 2015 SP4, Revision Number 23.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

Direction2 Property (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : Direction2 Property (ICircularPatternFeatureData)

Gets or sets whether to create a bidirectional circular pattern feature.

.NET Syntax

Property Direction2 As System.Boolean



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean

instance.Direction2 = value

value = instance.Direction2
```

```
System.bool Direction2 {get; set;}
```

```
property System.bool Direction2 {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to create a bidirectional circular pattern feature, false to not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::Direction2](#).

Example

[Create Bidirectional Circular Pattern Feature \(C#\)](#)
[Create Bidirectional Circular Pattern Feature \(VB.NET\)](#)
[Create Bidirectional Circular Pattern Feature \(VBA\)](#)

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::EqualSpacing2 Property \(\)](#)
[ICircularPatternFeatureData::Spacing2 Property \(\)](#)
[ICircularPatternFeatureData::Symmetric Property \(\)](#)
[ICircularPatternFeatureData::TotalInstances2 Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

EqualSpacing Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : EqualSpacing Property (ICircularPatternFeatureData)

Gets or sets whether the pattern instances in Direction 1 are equally spaced in this circular pattern feature.

.NET Syntax

Property EqualSpacing As System.Boolean

```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean

instance.EqualSpacing = value

value = instance.EqualSpacing
```

C#

```
System.bool EqualSpacing {get; set;}
```

C++/CLI

```
property System.bool EqualSpacing {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use equal spacing in Direction 1, false to not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::EqualSpacing](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::TotalInstances Property](#)
[ICircularPatternFeatureData::Spacing Property](#)
[ICircularPatternFeatureData::EqualSpacing2 Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

EqualSpacing2 Property (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : EqualSpacing2 Property (ICircularPatternFeatureData)

Gets or sets whether the pattern instances in Direction 2 are equally spaced in this bidirectional circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property EqualSpacing2 As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean

instance.EqualSpacing2 = value

value = instance.EqualSpacing2
```

C#

```
System.bool EqualSpacing2 {get; set;}
```

C++/CLI

```
property System.bool EqualSpacing2 {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use equal spacing in Direction 2, false to not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::EqualSpacing2](#).

Example

[Create Bidirectional Circular Pattern Feature \(C#\)](#)
[Create Bidirectional Circular Pattern Feature \(VB.NET\)](#)
[Create Bidirectional Circular Pattern Feature \(VBA\)](#)

Remarks

This property is only available when:

- [ICircularPatternFeatureData::Direction2](#) is true and
- [ICircularPatternFeatureData::Symmetric](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::Spacing2 Property \(\)](#)
[ICircularPatternFeatureData::TotalInstances2 Property \(\)](#)
[ICircularPatternFeatureData::EqualSpacing Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureScope Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : FeatureScope Property (ICircularPatternFeatureData)

Gets which bodies in this multibody part are affected by this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureScope As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Boolean
```

```
value = instance.FeatureScope
```

C#

```
System.bool FeatureScope {get;}
```

C++/CLI

```
property System.bool FeatureScope {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to specify affected bodies, false to apply the pattern to all bodies every time the feature regenerates

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::FeatureScope](#).

Remarks

This property is valid only if [ICircularPatternFeatureData::GeometryPattern](#) is true.

Call [ICircularPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Circular Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureScopeBodies Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : FeatureScopeBodies Property (ICircularPatternFeatureData)

Gets the bodies in this multibody part that are affected by this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureScopeBodies As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object  
  
value = instance.FeatureScopeBodies
```

C#

```
System.Object FeatureScopeBodies {get;}
```

C++/CLI

```
property System.Object^ FeatureScopeBodies {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) to be affected

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::FeatureScopeBodies](#).

Remarks

This property is valid only if [ICircularPatternFeatureData::GeometryPattern](#) is true.

Call [ICircularPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Circular Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GeometryPattern Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GeometryPattern Property (ICircularPatternFeatureData)

Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.

.NET Syntax

Visual Basic (Declaration)

```
Property GeometryPattern As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Boolean
```

```
instance.GeometryPattern = value
```

```
value = instance.GeometryPattern
```

C#

```
System.bool GeometryPattern {get; set;}
```

C++/CLI

```
property System.bool GeometryPattern {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use only the geometry of the feature to create the pattern (faster), false to pattern and solve each instance of the feature in its entirety (slower)

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GeometryPattern](#).

Remarks

This property is valid only if [ICircularPatternFeatureData::BodyPattern](#) is false.

If this property is set to true, you can call [ICircularPatternFeatureData::SetFeatureScope](#).

For more information, see the **Circular Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAxisType Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetAxisType Method (ICircularPatternFeatureData)

Gets the type of geometry used to determine the direction of this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAxisType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetAxisType()
```

C#

```
System.int GetAxisType()
```

C++/CLI

```
System.int GetAxisType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Geometry type used to create the circular pattern:

- 0 = reference axis
- 1 = edge
- 2 = dimension
- 3 = sketch line

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GetAxisType](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::Axis Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetInstanceToVaryOptions Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetInstanceToVaryOptions Method (ICircularPatternFeatureData)

Gets the options for varying the spacing of pattern instances in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetInstanceToVaryOptions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object  
  
value = instance.GetInstanceToVaryOptions()
```

C#

```
System.object GetInstanceToVaryOptions()
```

C++/CLI

```
System.Object^ GetInstanceToVaryOptions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IInstanceToVaryOptions](#)

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData.GetInstanceToVaryOptions](#).

Remarks

This method is valid only if [ICircularPatternFeatureData::InstancesToVary](#) is true.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::SetInstanceToVaryOptions Method \(\)](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternBodyCount Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetPatternBodyCount Method (ICircularPatternFeatureData)

Gets the number of seed bodies in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternBodyCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetPatternBodyCount()
```

C#

```
System.int GetPatternBodyCount()
```

C++/CLI

```
System.int GetPatternBodyCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed bodies

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GetPatternBodyCount](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternBodyArray Method](#)
[ICircularPatternFeatureData::PatternBodyArray Property](#)
[ICircularPatternFeatureArray::SetPatternBodyArray Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternFaceCount Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetPatternFaceCount Method (ICircularPatternFeatureData)

Gets the number of patterned faces.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternFaceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPatternFaceCount()
```

C#

```
System.int GetPatternFaceCount()
```

C++/CLI

```
System.int GetPatternFaceCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of patterned faces

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GetPatternFaceCount](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::IGetPatternFaceArray Method](#)
[ICircularPatternFeatureData::ISetPatternFaceArray Method](#)
[ICircularPatternFeatureData::PatternFaceArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternFeatureCount Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetPatternFeatureCount Method (ICircularPatternFeatureData)

Gets the number of seed features used to create this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternFeatureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetPatternFeatureCount()
```

C#

```
System.int GetPatternFeatureCount()
```

C++/CLI

```
System.int GetPatternFeatureCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed features

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GetPatternFeatureCount](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::IGetPatternFeatureArray Method](#)
[ICircularPatternFeatureData::ISetPatternFeatureArray Method](#)
[ICircularPatternFeatureData::PatternFeatureArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSkippedItemCount Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetSkippedItemCount Method (ICircularPatternFeatureData)

Gets the number of items skipped in this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSkippedItemCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetSkippedItemCount()
```

C#

```
System.int GetSkippedItemCount()
```

C++/CLI

```
System.int GetSkippedItemCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of skipped items

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GetSkippedItemCount](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::IGetSkippedItemArray Method](#)
[ICircularPatternFeatureData::ISetSkippedItemArray Method](#)
[ICircularPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTransform Method (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : GetTransform Method (ICircularPatternFeatureData)

Gets the transform for the specified instance of this circular-pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTransform( _  
    ByVal Instance As System.Integer _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim Instance As System.Integer  
Dim value As MathTransform
```

```
value = instance.GetTransform(Instance)
```

C#

```
MathTransform GetTransform(  
    System.int Instance  
)
```

C++/CLI

```
MathTransform^ GetTransform(  
& System.int Instance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Index of one repeating element in this pattern (see **Remarks**)

Return Value

[Transform](#)

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::GetTransform](#).

Example

[Get Transform for Each Circular Pattern Instance \(C#\)](#)
[Get Transform for Each Circular Pattern Instance \(VB.NET\)](#)
[Get Transform for Each Circular Pattern Instance \(VBA\)](#)

Remarks

1 <= *Instance* <= [ICircularPatternFeatureData::TotalInstances](#)

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAccessSelections Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : IAccessSelections Method (ICircularPatternFeatureData)

Obsolete. Superseded by [ICircularPatternFeatureData::IAccessSelections2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections( _  
    ByVal TopDoc As ModelDoc, _  
    ByVal Component As Component _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim TopDoc As ModelDoc
Dim Component As Component
Dim value As System.Boolean

value = instance.IAccessSelections(TopDoc, Component)
```

C#

```
System.bool IAccessSelections(
    ModelDoc TopDoc,
    Component Component
)
```

C++/CLI

```
System.bool IAccessSelections(
    & ModelDoc^ TopDoc,
    & Component^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Component

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::IAccessSelections](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAccessSelections2 Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : IAccessSelections2 Method (ICircularPatternFeatureData)

Gains access to selections used to define a circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections2( _
    ByVal TopDoc As ModelDoc2, _
    ByVal Component As Component2 _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim TopDoc As ModelDoc2
Dim Component As Component2
Dim value As System.Boolean
```

```
value = instance.IAccessSelections2(TopDoc, Component)
```

C#

```
System.bool IAccessSelections2(
    ModelDoc2 TopDoc,
```

```
        Component2 Component
    )
```

C++/CLI

```
System.bool IAccessSelections2(
&    ModelDoc2^ TopDoc,
&    Component2^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document

Component

Component for the feature

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::IAccessSelections2](#).

Remarks

Remark

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::IModifyDefinition2](#) if you modified the feature
- [ICircularPatternFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 FCS Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPatternBodyArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : IGetPatternBodyArray Method (ICircularPatternFeatureData)

Gets a list of the seed bodies for this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternBodyArray() As Body2
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As Body2
```

```
value = instance.IGetPatternBodyArray()
```

C#

```
Body2 IGetPatternBodyArray()
```

C++/CLI

```
Body2^ IGetPatternBodyArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of seed [bodies](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternBodyCount Method](#)
[ICircularPatternFeatureData::ISetPatternBodyArray Method](#)
[ICircularPatternFeatureData::PatternBodyArray Property](#)
[ICircularPatternFeatureData::BodyPattern](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPatternFaceArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : IGetPatternFaceArray Method (ICircularPatternFeatureData)

Gets a list of patterned faces for this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternFaceArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object
```

```
value = instance.IGetPatternFaceArray()
```

C#

```
System.object IGetPatternFaceArray()
```

C++/CLI

```
System.Object^ IGetPatternFaceArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of patterned [faces](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternFaceCount Method](#)
[ICircularPatternFeatureData::ISetPatternFaceArray Method](#)
[ICircularPatternFeatureData::PatternFaceArray Property](#)
[ICircularPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPatternFeatureArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : IGetPatternFeatureArray Method (ICircularPatternFeatureData)

Gets the seed features for this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternFeatureArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object  
  
value = instance.IGetPatternFeatureArray()
```

C#

```
System.Object IGetPatternFeatureArray()
```

C++/CLI

```
System.Object^ IGetPatternFeatureArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

in-process, unmanaged C++: Pointer to an array of seed [features](#)

- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::ISetPatternFeatureArray Method](#)
[ICircularPatternFeatureData::PatternFeatureArray Property](#)
[ICircularPatternFeatureData::GetPatternFeatureCount Method](#)
[ICircularPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSkippedItemArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : IGetSkippedItemArray Method (ICircularPatternFeatureData)

Gets an array of integers that represent the skipped items in this circular feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSkippedItemArray() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.IGetSkippedItemArray()
```

C#

```
System.int IGetSkippedItemArray()
```

C++/CLI

```
System.int IGetSkippedItemArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of integers that represent the skipped items in the pattern
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This is a 0-based array, so if you skip the third and fifth items in the pattern, then:

- Array[0] = 3
- Array[1] = 5

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetSkippedItemCount Method](#)
[ICircularPatternFeatureData::ISetSkippedItemArray Method](#)
[ICircularPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InstancesToVary Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : InstancesToVary Property (ICircularPatternFeatureData)

Gets or sets whether to vary the spacing of pattern instances in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property InstancesToVary As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean

instance.InstancesToVary = value

value = instance.InstancesToVary
```

C#

```
System.bool InstancesToVary {get; set;}
```

C++/CLI

```
property System.bool InstancesToVary {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to vary the spacing of pattern instances, false to not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::InstancesToVary](#).

Remarks

This property is valid only when editing the circular pattern feature. You cannot set this property during creation of the feature.

To vary instances in a circular pattern feature:

1. Call [IFeature::GetDefinition](#).
2. Call [ICircularPatternFeatureData::AccessSelections](#).
3. Set this property to true.
4. Call [IFeature::ModifyDefinition](#).
5. Call [ICircularPatternFeatureData::AccessSelections](#).
6. Call [ICircularPatternFeatureData::GetInstanceToVaryOptions](#).
7. Modify [InstanceToVaryOptions](#) properties.
8. Call [ICircularPatternFeatureData::SetInstanceToVaryOptions](#).
9. Call [IFeature::ModifyDefinition](#).

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternBodyArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : ISetPatternBodyArray Method (ICircularPatternFeatureData)

Sets a list of seed bodies for the circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternBodyArray( _
    ByVal BodyCount As System.Integer, _
    ByRef ArrayDataIn As Body2 _
)
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim BodyCount As System.Integer
Dim ArrayDataIn As Body2

instance.ISetPatternBodyArray(BodyCount, ArrayDataIn)
```

C#

```
void ISetPatternBodyArray(
    System.int BodyCount,
    ref Body2 ArrayDataIn
)
```

C++/CLI

```
void ISetPatternBodyArray(
    & System.int BodyCount,
    & Body2^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyCount

Number of seed bodies

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of seed [bodies](#) of size BodyCount
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternBodyCount Method](#)
[ICircularPatternFeatureData::GetPatternBodyArray Method](#)
[ICircularPatternFeatureData::PatternBodyArray Property](#)
[ICircularPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternFaceArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : ISetPatternFaceArray Method (ICircularPatternFeatureData)

Sets a list of patterned faces for this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternFaceArray( _
    ByVal FaceCount As System.Integer, _
    ByRef ArrayDataIn As System.Object _
)
```

Visual Basic (Usage)

```
Dim instance As ICircularPatternFeatureData
Dim FaceCount As System.Integer
Dim ArrayDataIn As System.Object
```



```
instance.ISetPatternFaceArray(FaceCount, ArrayDataIn)
```

C#

```
void ISetPatternFaceArray(  
    System.int FaceCount,  
    ref System.object ArrayDataIn  
)
```

C++/CLI

```
void ISetPatternFaceArray(  
& System.int FaceCount,  
& System.Object^% ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceCount

Number of patterned faces

ArrayDataIn

in-process, unmanaged C++: Pointer to an array of patterned [faces](#) of size FaceCount

- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)

[ICircularPatternFeatureData Members](#)

[ICircularPatternFeatureData::GetPatternFaceCount Method](#)

[ICircularPatternFeatureData::IGetPatternFaceArray Method](#)

[ICircularPatternFeatureData::PatternFaceArray Property](#)

[ICircularPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternFeatureArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : ISetPatternFeatureArray Method (ICircularPatternFeatureData)

Sets the seed features to use to create the circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternFeatureArray( _  
    ByVal FeatCount As System.Integer, _  
    ByRef ArrayDataIn As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim FeatCount As System.Integer  
Dim ArrayDataIn As System.Object
```

```
instance.ISetPatternFeatureArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetPatternFeatureArray(
    System.int FeatCount,
    ref System.object ArrayDataIn
)
```

C++/CLI

```
void ISetPatternFeatureArray(
    & System.int FeatCount,
    & System.Object^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of seed features to use to create this circular pattern

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of seed [features](#) to use to create the circular pattern

VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternFeatureCount Method](#)
[ICircularPatternFeatureData::GetPatternFeatureArray Method](#)
[ICircularPatternFeatureData::PatternFeatureArray Property](#)
[ICircularPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISetSkippedItemArray Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : ISetSkippedItemArray Method (ICircularPatternFeatureData)

Sets the list of skipped items in this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSkippedItemArray( _
    ByVal FeatCount As System.Integer, _
    ByRef ArrayDataIn As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim FeatCount As System.Integer
Dim ArrayDataIn As System.Integer

instance.ISetSkippedItemArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetSkippedItemArray(
    System.int FeatCount,
```

```
ref System.int ArrayDataIn
)
```

C++/CLI

```
void ISetSkippedItemArray(
& System.int FeatCount,
& System.int% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of instances to skip

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array that represents the skipped items of size FeatCount

• VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICircularPatternFeatureData Interface](#)

[ICircularPatternFeatureData Members](#)

[ICircularPatternFeatureData::GetSkippedItemCount Method](#)

[ICircularPatternFeatureData::IGetSkippedItemArray Method](#)

[ICircularPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternBodyArray Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : PatternBodyArray Property (ICircularPatternFeatureData)

Gets or sets a list of bodies to pattern in a multibody part for this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternBodyArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Object
```

```
instance.PatternBodyArray = value
```

```
value = instance.PatternBodyArray
```

C#

```
System.object PatternBodyArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternBodyArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) to pattern

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::PatternBodyArray](#).

Remarks

This property is valid only if:

- you have a multibody part,
- and -
- [ICircularPatternFeatureData::BodyPattern](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternBodyCount Method](#)
[ICircularPatternFeatureData::IGetPatternBodyArray Method](#)
[ICircularPatternFeatureData::ISetPatternBodyArray Method](#)
[ICircularPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternElement Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : PatternElement Property (ICircularPatternFeatureData)

Gets or sets the type of entities on which to base this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PatternElement As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer
```

```
instance.PatternElement = value
```

```
value = instance.PatternElement
```

C#

```
System.int PatternElement {get; set;}
```

C++/CLI

```
property System.int PatternElement {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of entities on which to base this circular pattern feature defined in [swPatternElementSelection_e](#)

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::PatternElement](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::BodyPattern Property \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternFaceArray Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : PatternFaceArray Property (ICircularPatternFeatureData)

Gets or sets the list of faces to pattern for this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternFaceArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Object

instance.PatternFaceArray = value
value = instance.PatternFaceArray
```

C#

```
System.object PatternFaceArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFaceArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [faces](#) to pattern

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::PatternFaceArray](#).

Remarks

This property is valid only if [ICircularPatternFeatureData::BodyPattern](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

[ICircularPatternFeatureData::GetPatternFaceCount Method](#)
[ICircularPatternFeatureData::IGetPatternFaceArray Method](#)
[ICircularPatternFeatureData::ISetPatternFaceArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternFeatureArray Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : PatternFeatureArray Property (ICircularPatternFeatureData)

Gets or sets the seed features for the circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternFeatureArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternFeatureArray = value
```

```
value = instance.PatternFeatureArray
```

C#

```
System.Object PatternFeatureArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFeatureArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [features](#) used to create this circular pattern feature

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::PatternFeatureArray](#).

Remarks

This property is valid only if [ICircularPatternFeatureData::BodyPattern](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetPatternFeatureCount Method](#)
[ICircularPatternFeatureData::IGetPatternFeatureArray Method](#)
[ICircularPatternFeatureData::ISetPatternFeatureArray Method](#)
[ICircularPatternFeatureData::InstancesToVary Property\(\)](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PropagateVisualProperty Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : PropagateVisualProperty Property (ICircularPatternFeatureData)

Gets or sets whether to propagate visual properties (e.g., colors, textures, cosmetic thread data, etc.) to all pattern instances in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PropagateVisualProperty As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Boolean
```

```
instance.PropagateVisualProperty = value
```

```
value = instance.PropagateVisualProperty
```

C#

```
System.bool PropagateVisualProperty {get; set;}
```

C++/CLI

```
property System.bool PropagateVisualProperty {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to propagate all visual properties, false to not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::PropagateVisualProperty](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : ReleaseSelectionAccess Method (ICircularPatternFeatureData)

Releases access to the selections that define this circular pattern.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

```
Dim instance As ICircularPatternFeatureData

instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::ReleaseSelectionAccess](#).

Remarks

[ICircularPatternFeatureData::AccessSelections](#) or [ICircularPatternFeatureData::!AccessSelections2](#) puts the model into a rollback state to allow access to the selections that define this feature.

Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) or [IFeature::!ModifyDefinition2](#) if you did.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReverseDirection Property (ICircularPatternFeatureData)

[See Also](#)


[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : ReverseDirection Property (ICircularPatternFeatureData)

Gets or sets whether the direction of the axis should be reversed in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ReverseDirection As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ICircularPatternFeatureData
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True reverses the direction, false does not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::ReverseDirection](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetFeatureScope Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : SetFeatureScope Method (ICircularPatternFeatureData)

Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFeatureScope( _  
    ByVal FeatureScopeOption As System.Boolean, _  
    ByVal AutoSelectBodies As System.Boolean, _  
    ByVal Bodies As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim FeatureScopeOption As System.Boolean  
Dim AutoSelectBodies As System.Boolean  
Dim Bodies As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetFeatureScope(FeatureScopeOption, AutoSelectBodies, Bodies)
```

C#

```
System.bool SetFeatureScope(  
    System.bool FeatureScopeOption,  
    System.bool AutoSelectBodies,  
    System.object Bodies  
)
```

C++/CLI

```
System.bool SetFeatureScope(  
& System.bool FeatureScopeOption,  
& System.bool AutoSelectBodies,  
& System.Object^ Bodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureScopeOption

True (default) to specify affected bodies, false to apply the pattern to all bodies every time the feature regenerates (see **Remarks**)

AutoSelectBodies
True (default) to automatically select all bodies intersected by this pattern feature, false to specify affected bodies (see **Remarks**)
Bodies
Array of [bodies](#) to be affected; valid only if FeatureScopeOption is true and AutoSelectBodies is false

Return Value

True if feature scope set successfully, false if not (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::SetFeatureScope](#).

Remarks

If this method returns false, then default values for FeatureScopeOption and AutoSelectBodies are set. A subsequent call to [IFeatureManager::CreateFeature](#) creates the feature with a FeatureScopeOption of true and an AutoSelectBodies of true.

After calling IFeature::ModifyDefinition, call [IFeature::GetErrorCode2](#) to determine what's wrong and then take necessary remedial action.

For more information, see the **Circular Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::AutoSelect Property \(\)](#)
[ICircularPatternFeatureData::FeatureScope Property \(\)](#)
[ICircularPatternFeatureData::FeatureScopeBodies Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetInstanceToVaryOptions Method (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : SetInstanceToVaryOptions Method (ICircularPatternFeatureData)

Sets the options for varying the spacing of pattern instances in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetInstanceToVaryOptions( _  
    ByVal InstancesToVaryObj As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim InstancesToVaryObj As System.Object  
Dim value As System.Boolean  
  
value = instance.SetInstanceToVaryOptions(InstancesToVaryObj)
```

C#

```
System.bool SetInstanceToVaryOptions(  
    System.object InstancesToVaryObj  
)
```

C++/CLI

```
System.bool SetInstanceToVaryOptions(  
& System.Object^ InstancesToVaryObj  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InstancesToVaryObj
[InstancesToVaryOptions](#)

Return Value

True if options successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::SetInstanceToVaryOptions](#).

Remarks

This method is valid only if [ICircularPatternFeatureData::InstancesToVary](#) is true.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetInstanceToVaryOptions Method \(\)](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments](#) on this topic.

SkippedItemArray Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : SkippedItemArray Property (ICircularPatternFeatureData)

Gets or sets the list of items to skip in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property SkippedItemArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Object
```

```
instance.SkippedItemArray = value
```

```
value = instance.SkippedItemArray
```

C#

```
System.object SkippedItemArray {get; set;}
```

C++/CLI

```
property System.Object^ SkippedItemArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of integers representing the items to skip

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::SkippedItemArray](#).

Remarks

This array is 0-based. If you skip the third and fifth items, then the array looks like `ArrayOut[0]=3` and `ArrayOut[1]=5`.

To calculate the pattern instance numbers to skip:

Calculate:

For a bi-directional pattern:

$$I = n2 * (i - 1) + (j - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- I = pattern instance number
- $n2$ = number of instances in Direction 2
- i = index for Direction 1
- j = index for Direction 2

In the pattern's PropertyManager, find $n2$ in the **Direction 2 > Spacing and Instances > Number of instances** field and find $[i,j]$ in the **Modified Instances** section.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::GetSkippedItemCount Method](#)
[ICircularPatternFeatureData::GetSkippedItemArray Method](#)
[ICircularPatternFeatureData::SetSkippedItemArray Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Spacing Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : Spacing Property (ICircularPatternFeatureData)

Gets or sets the spacing between pattern instances in Direction 1 of the circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property Spacing As System.Double

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Double
```

```
instance.Spacing = value
```

```
value = instance.Spacing
```

C#

```
System.double Spacing {get; set;}
```

C++/CLI

```
property System.double Spacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance in radians between pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::Spacing](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::EqualSpacing Property](#)
[ICircularPatternFeatureData::TotalInstances Property](#)
[ICircularPatternFeatureData::Spacing2 Property](#)
[ICircularPatternFeatureData::InstancesToVary Property\(\)](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Spacing2 Property (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : Spacing2 Property (ICircularPatternFeatureData)

Gets or sets the spacing between pattern instances in Direction 2 of this bidirectional circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property Spacing2 As System.Double

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Double
```

```
instance.Spacing2 = value  
value = instance.Spacing2
```

C#

```
System.double Spacing2 {get; set;}
```

C++/CLI

```
property System.double Spacing2 {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance in radians between pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::Spacing2](#).

Example

[Create Bidirectional Circular Pattern Feature \(C#\)](#)
[Create Bidirectional Circular Pattern Feature \(VB.NET\)](#)
[Create Bidirectional Circular Pattern Feature \(VBA\)](#)

Remarks

This property is only available when:

- [ICircularPatternFeatureData::Direction2](#) is true, and
- [ICircularPatternFeatureData::Symmetric](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::EqualSpacing2 Property](#)
[ICircularPatternFeatureData::TotalInstances2 Property \(\)](#)
[ICircularPatternFeatureData::Spacing Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

StructureSystemToPatternArray Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : StructureSystemToPatternArray Property (ICircularPatternFeatureData)

Gets or sets the structure systems to pattern in this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property StructureSystemToPatternArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData
Dim value As System.Object

instance.StructureSystemToPatternArray = value

value = instance.StructureSystemToPatternArray
```

C#

```
System.object StructureSystemToPatternArray {get; set;}
```

C++/CLI

```
property System.Object^ StructureSystemToPatternArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [IStructureSystemFolders](#)

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::StructureSystemToPatternArray](#).

Example

VBA

This example demonstrates how to create a circular pattern using a structure system.

```

'=====
'Preconditions:
'1. Ensure the specified part template exists.
'2. Open an Immediate window.
'3. Press F5.
'
'Postconditions:
' 1. Creates Sketch1 containing two sketch segments.
' 2. Configures the start/end extensions and the member profile.
' 3. Selects the two sketch segments.
' 4. Creates primary Structure System1 with two primary path segment members (Member1 and Member2).
' 5. Creates an axis about which to create a circular pattern.
' 6. Creates a Boss-Extrude1 to pattern.
' 7. Inspect the graphics area.
' 8. Press F5 to create CirPattern1.
' 9. Press F5 to finish.
'10. Inspect the Immediate window.
'=====
Dim swApp As SldWorks.SldWorks
Dim modDoc As SldWorks.ModelDoc2
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim modDocExt As SldWorks.ModelDocExtension
Dim structMemDef As SldWorks.StructureSystemMemberFeatureData
Dim profDef As SldWorks.StructureSystemMemberProfile
Dim PrimDef As SldWorks.PrimaryStructuralMemberFeatureData
Dim memPathSegDef As PrimaryMemberPathSegmentFeatureData
Dim memBetweenPointsSecDef As SldWorks.SecondaryMemberBetweenPointsFeatureData
Dim secMemDatToChange As SldWorks.SecondaryMemberBetweenPointsFeatureData
Dim memSupportPlaneSecDef As SldWorks.SecondaryMemberSupportPlaneFeatureData
Dim segments(2) As Object
Dim stat As Boolean
Dim memData(0) As SldWorks.StructureSystemMemberFeatureData
Dim memDataArray As Variant
Dim structSys As SldWorks.Feature
Dim baseSecondarySystem As SldWorks.StructureSystemFolder
Dim structSysDef As SldWorks.StructureSystemFolder
Dim structSysSecDef As SldWorks.StructureSystemFolder
Dim feat As SldWorks.Feature
Dim structSysModDef As SldWorks.StructureSystemFolder
Dim outProfiles As Variant
Dim MemberData As SldWorks.StructureSystemMemberFeatureData
Dim memberArray As Variant
Dim profileGrpFeat As SldWorks.Feature
Dim profileGrp As SldWorks.ProfileGroupFolder
Dim memToChange As SldWorks.StructureSystemMemberFeatureData
Dim i As Long
Dim j As Long
Dim secDef As SldWorks.SecondaryStructuralMemberFeatureData
Dim UpToMemDef As SldWorks.SecondaryMemberUpToMembersFeatureData
Dim point As Object
Dim Members(0) As Object

Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchMgr As SldWorks.SketchManager
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSelectionMgr As SldWorks.SelectionMgr
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swCircularPatternFeatureData As SldWorks.CircularPatternFeatureData
Dim sketchSegments As Variant
Dim status As Boolean
Dim obj As Object
Dim patternFeatures(1) As Object

Option Explicit

Sub main()
    Set swApp = Application.SldWorks

    Dim swSheetWidth As Double
    swSheetWidth = 0
    Dim swSheetHeight As Double
    swSheetHeight = 0
    Set modDoc = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2023\templates\Part.PRTDOT", 0, swSheetWidth,
swSheetHeight)

    Set modDocExt = modDoc.Extension

    modDoc.SketchManager.InsertSketch True
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", -0.077188654347454, 0.054268560279924, 3.86214196026222E-03, False, 0,

```

```

Nothing, 0)
modDoc.ClearSelection2 True

Dim skSegment As Object
Set skSegment = modDoc.SketchManager.CreateLine(-0.168061, 0.084736, 0#, -0.168061, -0.077684, 0#)
Set skSegment = modDoc.SketchManager.CreateLine(0.075216, 0.107771, 0#, 0.075216, -0.006699, 0#)
modDoc.ClearSelection2 True

stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
Dim myRefPlane As SldWorks.RefPlane
Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(8, 0.03, 0, 0, 0, 0)
modDoc.ClearSelection2 True
stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
modDoc.ClearSelection2 True
modDoc.SketchManager.InsertSketch True

stat = modDocExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.168061313304597, 5.44142573846353E-02, 0, False, 0,
Nothing, 0)
stat = modDocExt.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", 7.52162521083513E-02, 5.32390034454423E-02, 0, True, 0,
Nothing, 0)

'Create primary structure system path segment members
Set structMemDef = modDocExt.CreateStructureSystemMemberData(0)
Debug.Print "Type of structure system member as defined by swStructureSystemMemberType_e: " &
structMemDef.StructureSystemMemberType

structMemDef.StartEndExtendD1 = 1#
structMemDef.StartEndExtendD2 = 2#

Set profDef = structMemDef.MemberProfile

profDef.ProfileStandard = "ansi inch"
profDef.ProfileType = "c channel"
profDef.ProfileSize = "3 x 5"

Set PrimDef = structMemDef
Debug.Print "Structure system primary member creation type as defined by swStructureSystemMemberCreationType_e: " &
PrimDef.PrimaryMemberType

Set memPathSegDef = PrimDef
memPathSegDef.MergeTangentMembers = True

Set swSelMgr = modDoc.SelectionManager

Dim segments(1) As Object
Set segments(0) = swSelMgr.GetSelectedObject6(1, 0)
Set segments(1) = swSelMgr.GetSelectedObject6(2, 0)
stat = memPathSegDef.SetPathSegments(segments)
Debug.Print "Path segments successfully set: " & stat

Dim PrimMemData(0) As SldWorks.StructureSystemMemberFeatureData
Set PrimMemData(0) = structMemDef

Dim PrimMemDatArray As Variant
PrimMemDatArray = PrimMemData

Dim structSysFeat As SldWorks.Feature
Set structSysFeat = modDocExt.CreateStructureSystem(PrimMemDatArray, Nothing)

'Create the axis for the circular pattern
status = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
modDoc.SketchManager.InsertSketch True
modDoc.ClearSelection2 True
Dim skSegmentAxis As Object
Set skSegmentAxis = modDoc.SketchManager.CreateLine(0.695357, 0.586267, 0#, 0.695357, -0.704287, 0#)
modDoc.ClearSelection2 True
modDoc.SketchManager.InsertSketch True

'Create a solid body to also pattern
status = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
modDoc.SketchManager.InsertSketch True
modDoc.ClearSelection2 True
status = modDocExt.SketchBoxSelect("0.509957", "-0.990370", "0.000000", "1.024907", "-1.314598", "0.000000")
status = modDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
status = modDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
Dim vSkLines As Variant
vSkLines = modDoc.SketchManager.CreateCornerRectangle(0.713394086918119, -0.907724114572968, 0, 1.44449597678115,
-1.33367043301491, 0)
modDoc.ClearSelection2 True

```

```

modDoc.SketchManager.InsertSketch True
modDoc.SketchManager.InsertSketch True
Dim myFeature As SldWorks.Feature
Set myFeature = modDoc.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0, 0.07, 0.01, False, False, False, False,
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, False, True, True, 0, 0, False)
modDoc.SelectionManager.EnableContourSelection = False

modDoc.ViewZoomtofit

Stop

Set swModel = swApp.ActiveDoc
Set swModelDocExt = swModel.Extension
Set swSketchMgr = swModel.SketchManager
Set swFeatureMgr = swModel.FeatureManager
Set swSelectionMgr = swModel.SelectionManager

'Pre-select axis/direction, bodies, structure system folders
status = swModelDocExt.SelectByID2("Line1@Sketch12", "EXTSKETCHSEGMENT", 0.695357, 0.185750016352603, 0, False, 1, Nothing, 0
status = swModelDocExt.SelectByID2("Structure System1", "BODYFEATURE", 0, 0, 0, True, 134217728, Nothing, 0)
status = swModelDocExt.SelectByID2("Boss-Extrude1", "SOLIDBODY", 0, 0, 0, True, 256, Nothing, 0)

'Create circular pattern
Set swCircularPatternFeatureData = swFeatureMgr.CreateDefinition(swFmCirPattern)
swCircularPatternFeatureData.EqualSpacing = True
swCircularPatternFeatureData.ReverseDirection = True
swCircularPatternFeatureData.Spacing = 1.919862 '110 degrees
swCircularPatternFeatureData.TotalInstances = 3
swCircularPatternFeatureData.GeometryPattern = False
swCircularPatternFeatureData.VarySketch = False

'Set BodyPattern property to True to pattern structure system folders
swCircularPatternFeatureData.BodyPattern = True

Set swFeature = swFeatureMgr.CreateFeature(swCircularPatternFeatureData)

Stop 'Examine the graphics area

swModel.ClearSelection2 True
status = swModelDocExt.SelectByID2("CirPattern1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeature = swSelectionMgr.GetSelectedObject6(1, -1)
Set swCircularPatternFeatureData = swFeature.GetDefinition
swCircularPatternFeatureData.AccessSelections swModel, Nothing
Debug.Print "Is CirPattern1 a features/faces pattern or a bodies pattern (true if a bodies pattern)? " &
swCircularPatternFeatureData.BodyPattern

Dim var As Variant
var = swCircularPatternFeatureData.StructureSystemToPatternArray

Dim strctSysFolder As StructureSystemFolder
Dim index As Integer
For index = LBound(var) To UBound(var)
    Set strctSysFolder = var(index)
    Set patternFeatures(0) = strctSysFolder
    Debug.Print "Number of structure system folders: " & strctSysFolder.GetProfileGroupFoldersCount
Next

'Uncomment the following lines of code to modify the circular pattern
'to use another structure system (e.g., Structure System2)
'Stop

'Dim feat As Feature
'swModel.ClearSelection2 True
'status = swModelDocExt.SelectByID2("Structure System2", "BODYFEATURE", 0, 0, 0, True, 0, Nothing, 0)
'Set feat = swSelectionMgr.GetSelectedObject6(1, 0)
'Set patternFeatures(1) = feat.GetSpecificFeature2()

"During circular pattern modification, use StructureSystemPatternArray property to change structure system to pattern
'swCircularPatternFeatureData.StructureSystemToPatternArray = patternFeatures

Debug.Print swFeature.ModifyDefinition(swCircularPatternFeatureData, swModel, Nothing)

End Sub

```

Remarks

This property is valid only if [ICircularPatternFeatureData::BodyPattern](#) is true.

Use the setter of this property to change structure systems to pattern only after the feature has been created.

Before using this property in .NET code, read [Dispatch Object Arrays as Input in .NET](#). To remove a structure system from the pattern, set this property to a Dispatcher array containing one null element.

For more information, see the **SOLIDWORKS Design user-interface help > Weldments and Structure System > Pattern and Mirror Support** topic.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

Symmetric Property (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : Symmetric Property (ICircularPatternFeatureData)

Gets or sets whether to create a symmetric or asymmetric circular pattern feature in Direction 2.

.NET Syntax

Visual Basic (Declaration)

Property Symmetric As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Boolean
```

```
instance.Symmetric = value
```

```
value = instance.Symmetric
```

C#

```
System.bool Symmetric {get; set;}
```

C++/CLI

```
property System.bool Symmetric {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to create a symmetric circular pattern feature in Direction 2, false to create an asymmetrical circular pattern feature in Direction 2

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::Symmetric](#).

Example

[Create Bidirectional Circular Pattern Feature \(C#\)](#)
[Create Bidirectional Circular Pattern Feature \(VB.NET\)](#)
[Create Bidirectional Circular Pattern Feature \(VBA\)](#)

Remarks

This property is only available when [ICircularPatternFeatureData::Direction2](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData.EqualSpacing2 Property \(\)](#)
[ICircularPatternFeatureData.Spacing2 Property \(\)](#)
[ICircularPatternFeatureData.TotalInstances2 Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

TotalInstances Property (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : TotalInstances Property (ICircularPatternFeatureData)

Gets or sets the total number of pattern instances in Direction 1 of this circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property TotalInstances As System.Integer

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer
```

```
instance.TotalInstances = value
```

```
value = instance.TotalInstances
```

C#

```
System.int TotalInstances {get; set;}
```

C++/CLI

```
property System.int TotalInstances {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Total number of pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData.TotalInstances](#).

Example

[Get Transform for Each Circular Pattern Instance \(C#\)](#)
[Get Transform for Each Circular Pattern Instance \(VB.NET\)](#)
[Get Transform for Each Circular Pattern Instance \(VBA\)](#)

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData.EqualSpacing Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

TotalInstances2 Property (ICircularPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : TotalInstances2 Property (ICircularPatternFeatureData)

Gets or sets the total number of pattern instances in Direction 2 of this bidirectional circular pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property TotalInstances2 As System.Integer

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Integer
```

```
instance.TotalInstances2 = value
```

```
value = instance.TotalInstances2
```

C#

```
System.int TotalInstances2 {get; set;}
```

C++/CLI

```
property System.int TotalInstances2 {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Total number of pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::TotalInstances2](#).

Example

[Create Bidirectional Circular Pattern Feature \(C#\)](#)
[Create Bidirectional Circular Pattern Feature \(VB.NET\)](#)
[Create Bidirectional Circular Pattern Feature \(VBA\)](#)

Remarks

This property is only available when:

- [ICircularPatternFeatureData::Direction2](#) is true, and
- [ICircularPatternFeatureData::Symmetric](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)
[ICircularPatternFeatureData::EqualSpacing2 Property \(\)](#)
[ICircularPatternFeatureData::Spacing2 Property \(\)](#)
[ICircularPatternFeatureData::TotalInstances Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

VarySketch Property (ICircularPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICircularPatternFeatureData Interface](#) : VarySketch Property (ICircularPatternFeatureData)

Gets or sets whether to allow pattern instances of a seed sketch to vary in relation to the base part in this circular pattern.

.NET Syntax

Visual Basic (Declaration)

Property VarySketch As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICircularPatternFeatureData  
Dim value As System.Boolean
```

```
instance.VarySketch = value
```

```
value = instance.VarySketch
```

C#

```
System.bool VarySketch {get; set;}
```

C++/CLI

```
property System.bool VarySketch {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to allow the pattern instances of a seed sketch to vary in relation to enclosing geometry, false to not

Visual Basic for Applications (VBA) Syntax

See [CircularPatternFeatureData::VarySketch](#).

Remarks

This property is valid only when creating a pattern from a seed sketch.

The seed sketch has:

- An enclosing boundary of base geometry.
- Fully defined relations.
- Dimensions only for measurements that will not vary in the pattern instances.
- A driving dimension that is used to specify the direction of the circular pattern.

If this property is true:

- Specify [ICircularPatternFeatureData::Axis](#) with a driving dimension of the seed sketch.
- Pattern instances vary with and are constrained by the enclosing geometry.

See the **SOLIDWORKS Help > Parts and Features > Features > Patterns and Mirroring > Control and Modify Patterns > Vary Sketch** topic.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICircularPatternFeatureData Interface](#)
[ICircularPatternFeatureData Members](#)

Availability

